

CHARMM-GUI

Effective Simulation Input Generator and More

CHARMM is a versatile program for atomic-level simulation of many-particle systems, particularly macromolecules of biological interest. - M. Karplus

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Membrane Builder

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PDB Info	STEP 1	STEP 2	STEP 3	STEP 4	STEP 5	STEP 6	JOB ID: 3757172590
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Oriented PDB:	step2_orient.pdb (view structure)	download.tgz
System Size Input:	step3_size.inp	
System Size Output:	step3_size.out	
System Size:	step3_size.str	
Packing Simulation:	step3_packing.inp step3_packing.out crystal_image.str step3_packing_pol.str step3_packing.pdb (view structure)	Packing Simulation Input Packing Simulation Output Crystal Image Topology File of Pseudo Lipid Spheres Generated Packed System (please view this structure before you move to the next step)

Determined System Size:

Box Type	Rectangle		
Crystal Type	TETRAGONAL		
System Size	A	95.0202768	Dimension along the A (X) axis
	B	95.0202768	Dimension along the B (Y) axis
	C	95.4557052	Dimension along the C (Z) axis
Crystal Angle	Alpha	90.0	Angle between the axis B and C
	Beta	90.0	Angle between the axis A and C
	Gamma	90.0	Angle between the axis A and B
# of Lipids	on Top	115	
	on Bottom	118	
Z Center	1.0444318		Center of the system along the Z axis

System Building Options:

- ☐ Insertion method Build system using insertion method
☒ Replacement method Build system using replacement method
☒ Check lipid ring (and protein surface) penetration

For this system, insertion method can not be used. Replacement method will be used instead.

Component Building Options:

☒ Include IonsIon Placing Method: Distance ☒ Basic Ion TypesKCl 

Add Simple Ion Type

☐ More Ion Types**Formula Cation Anion Concentration Neutralizing**

KCl

 K^+ Cl^-

0.15



-

Calculate Solvent Composition

Ion	Count
K^+	48
Cl^-	37

Please note that the ion count is an approximation based on geometry. The real number will be calculated in the next step.

Pore Water Options:

The addition of pore water may result in water molecules placed outside the protein. Please download the structure file and open it using your molecular visualization software. In the list below, click water molecules that are outside the pore and should be removed.

Water filled protein structure: [step2.1_pore.pdb](#)

Check atom number of OH2 to remove

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20				
1	5195	5198	5201	5204	5207	6	5210	5213	5216	5219	5222	11	5225	5228	5231	5234	5237	16	5240	5243	5246	5249	5252
21	5255	5258	5261	5264	5267	26	5270	5273	5276	5279	5282	31	5285	5288	5291	5294	5297	36	5300	5303	5306	5309	5312

 Next Step: 
 Build Components

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